

MTH166

Lecture-7

Linear Differential Equations (LDE)

Unit 2: Differential Equations of Higher Order

(Book: Advanced Engineering Mathematics by R.K.Jain and S.R.K Iyengar, Chapter-5)

Topic:

Linear Differential Equations (LDE)

Learning Outcomes:

1. Identification of Linear Differential Equations (LDE).
2. Necessary and Sufficient condition for LDE to be Normal on an interval

Linear Differential Equations (LDE):

A linear differential equation of order n is written as:

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = r(x) \quad (1)$$

or

$$a_0 y^n + a_1 y^{n-1} + \dots + a_{n-1} y' + a_n y = r(x) \quad (1)$$

For example, a second order LDE is written as:

$$a_0 y'' + a_1 y' + a_2 y = r(x) \quad (2)$$

* If $r(x) = 0$, then LDE is called Homogeneous LDE.

* If $r(x) \neq 0$, then LDE is called Non-Homogeneous LDE.

* If $a_0, a_1, \dots, a_n$ are all constants, then LDE is called LDE with constant coefficients.

* If $a_0, a_1, \dots, a_n$ are not all constants, then it is called LDE with variable coefficients.

Classify the following LDE:

1. $y'' + 4y' + 3y = x^2 e^x$

It is a 2nd order Non-homogeneous LDE with constant coefficients.

2. $y'' + 2y' + y = \sin x$

It is a 2nd order Non-homogeneous LDE with constant coefficients.

3. $x^2 y'' + xy' + (x^2 - 4)y = 0$

It is a 2nd order Homogeneous LDE with variable coefficients.

4. $(1 - x^2)y'' - 2xy' + 20y = 0$

It is a 2nd order Homogeneous LDE with variable coefficients.

Polling Question

The equation: $y'' + 4y' + xy = x^2 e^x$ is:

- (A) 1st order Homogeneous LDE with variable coefficients.
- (B) 2nd order Homogeneous LDE with constant coefficients.
- (C) 2nd order Non-homogeneous LDE with variable coefficients.
- (D) 2nd order Non-homogeneous LDE with constant coefficients.

Necessary and Sufficient condition for LDE to be Normal on an interval:

A linear differential equation of order n :

$$a_0 y^n + a_1 y^{n-1} + \dots + a_{n-1} y' + a_n y = r(x) \quad (1)$$

is said to be normal on an interval I if:

1. $a_0, a_1, \dots, a_n$ and $r(x)$ are all continuous on an interval I

2. $a_0 \neq 0$

* In homogeneous LDE ($r(x)=0$), the problem arises only because of $a_0 \neq 0$.

* In non-homogeneous LDE ($r(x) \neq 0$), the problem arises due to $a_0 \neq 0$ and also due to domain of $r(x)$.

* In almost all numerical $a_0, a_1, \dots, a_n$ are all continuous, so first condition automatically holds. We are focused on second condition only.

Find the intervals on which the following differential equations are normal.

Problem 1. $(1 - x^2)y'' - 2xy' + 3y = 0$

Solution: $(1 - x^2)y'' - 2xy' + 3y = 0$ (1)

Comparing with: $a_0y'' + a_1y' + a_2y = 0$

$$\left. \begin{array}{l} a_0 = (1 - x^2) \\ 1. \ a_1 = -2x \\ \quad a_2 = 3 \end{array} \right\} \text{Being polynomials, } a_0, a_1, a_2 \text{ are all continuous on } (-\infty, \infty)$$

$$2. \ a_0 \neq 0 \quad \Rightarrow (1 - x^2) \neq 0 \quad \Rightarrow x^2 \neq 1 \quad \Rightarrow x \neq \pm 1$$



Thus, LDE (1) is normal on subintervals: $(-\infty, -1)$, $(-1, 1)$, $(1, \infty)$.

Problem 2. $x^2 y'' + xy' + (n^2 - x^2)y = 0$; n is real.

Solution: $x^2 y'' + xy' + (n^2 - x^2)y = 0$ (1)

Comparing with: $a_0 y'' + a_1 y' + a_2 y = 0$

$$1. \left. \begin{array}{l} a_0 = x^2 \\ a_1 = x \\ a_2 = (n^2 - x^2) \end{array} \right\} \text{Being polynomials, } a_0, a_1, a_2 \text{ are all continuous on } (-\infty, \infty)$$

$$2. a_0 \neq 0 \Rightarrow x^2 \neq 0 \Rightarrow x \neq 0$$



Thus, LDE (1) is normal on subintervals: $(-\infty, 0)$, $(0, \infty)$.

Problem 3. $x(1 - x)y'' - 3xy' - y = 0$

Solution: $x(1 - x)y'' - 3xy' - y = 0$ (1)

Comparing with: $a_0y'' + a_1y' + a_2y = 0$

$$\left. \begin{array}{l} a_0 = x(1 - x) \\ 1. \ a_1 = -3x \\ \quad a_2 = -1 \end{array} \right\} \text{Being polynomials, } a_0, a_1, a_2 \text{ are all continuous on } (-\infty, \infty)$$

$$2. \ a_0 \neq 0 \quad \Rightarrow \ x(1 - x) \neq 0 \quad \Rightarrow \ x \neq 0, \ x \neq 1$$



Thus, LDE (1) is normal on subintervals: $(-\infty, 0)$, $(0, 1)$, $(1, \infty)$.

Problem 4. $y'' + 9y' + y = \log(x^2 - 9)$

Solution: $y'' + 9y' + y = \log(x^2 - 9)$ (1)

Comparing with: $a_0y'' + a_1y' + a_2y = r(x)$

1. $\left. \begin{array}{l} a_0 = 1 \\ a_1 = 9 \\ a_2 = 1 \end{array} \right\}$ Being constants, a_0, a_1, a_2 are all continuous on $(-\infty, \infty)$

Also $r(x) = \log(x^2 - 9)$ will be defined if $(x^2 - 9) > 0$ i.e. $x^2 > 9$

$\Rightarrow |x| > 3 \quad \Rightarrow -\infty < x < -3$ and $3 < x < \infty$

2. $a_0 \neq 0 \quad \Rightarrow 1 \neq 0$ which is true.



Thus, LDE (1) is normal on subintervals: $(-\infty, -3), (3, \infty)$.

Problem 5. $\sqrt{x}y'' + 6xy' + 15y = \log(x^4 - 256)$

Solution: $\sqrt{x}y'' + 6xy' + 15y = \log(x^4 - 256)$ (1)

Comparing with: $a_0y'' + a_1y' + a_2y = r(x)$

1. $\left. \begin{array}{l} a_0 = \sqrt{x} \\ a_1 = 6x \\ a_2 = 15 \end{array} \right\} a_1, a_2 \text{ are all continuous on } (-\infty, \infty), \text{ but } a_0 \text{ is continuous on } (0, \infty)$

Also $r(x) = \log(x^4 - 256)$ will be defined if $(x^4 - 256) > 0$ i.e. $x^4 > 256$

$\Rightarrow |x| > 4 \Rightarrow -\infty < x < -4 \text{ and } 4 < x < \infty$

2. $a_0 \neq 0 \Rightarrow \sqrt{x} \neq 0 \Rightarrow x > 0$ (Square root of a negative number is not defined)



Thus, LDE (1) is normal on subintervals: $(4, \infty)$.

Polling Question

A linear differential equation: $a_0y'''' + a_1y''' + a_2y'' + a_3y' + a_4y = 0$ is said to be normal on an interval I if:

- (A) a_0, a_1, a_2, a_3 and $r(x)$ are all continuous on an interval I and $a_0 \neq 0$
- (B) a_0, a_1, a_2, a_3 are all continuous on an interval I and $a_0 \neq 0$
- (C) $a_0 \neq 0$

Problem 6. $y'' + 3y' + \sqrt{x}y = \sin x$

Solution: $y'' + 3y' + \sqrt{x}y = \sin x$ (1)

Comparing with: $a_0y'' + a_1y' + a_2y = r(x)$

$$1. \left. \begin{array}{l} a_0 = 1 \\ a_1 = 3 \\ a_2 = \sqrt{x} \end{array} \right\} a_0, a_1 \text{ are all continuous on } (-\infty, \infty), \text{ but } a_2 \text{ is continuous on } (0, \infty)$$

Also $r(x) = \sin x$ is continuous on $(-\infty, \infty)$.

2. $a_0 \neq 0 \Rightarrow 1 \neq 0$ which is true.



Thus, LDE (1) is normal on subintervals: $(0, \infty)$.

